TORQUE

DESCRIPTION	N·m	Ft. Lbs.	In. Lbs.
Camshaft Position Sensor Mounting Bolt – 4.0L	2	-	15
Camshaft Position Sensor – 2.4L	12	-	106
Ignition Coil Rail Mounting Bolts – 4.0L	29	-	250
Ignition Coil Mounting Bolts – 2.4L	11	-	105
Oil Pump Drive Hold-down Bolt – 4.0L	23	17	-
Oil Pump Drive Housing Cover Screws	-	-	-
** Spark Plugs - 2.4L 4-Cylinder	15	11	-
Spark Plugs - 4.0L 6-cylinder	30 +/- 5	22 +/- 4	-
** Torque critical tapered design. Do not exceed 15 ft. lbs. torque.			

AUTO SHUT DOWN RELAY

DESCRIPTION - PCM OUTPUT

The 5–pin, 12–volt, Automatic Shutdown (ASD) relay is located in the Power Distribution Center (PDC). Refer to label on PDC cover for relay location.

OPERATION

ASD SENSE - PCM INPUT

A 12 volt signal at this input indicates to the PCM that the ASD has been activated. The relay is used to connect the oxygen sensor heater element, ignition coil and fuel injectors to 12 volt + power supply.

This input is used only to sense that the ASD relay is energized. If the Powertrain Control Module (PCM) does not see 12 volts at this input when the ASD should be activated, it will set a Diagnostic Trouble Code (DTC).

PCM OUTPUT

The ASD relay supplies battery voltage (12+ volts) to the fuel injectors and ignition coil(s). With certain emissions packages it also supplies 12–volts to the oxygen sensor heating elements.

The ground circuit for the coil within the ASD relay is controlled by the Powertrain Control Module (PCM). The PCM operates the ASD relay by switching its ground circuit on and off.

The ASD relay will be shut–down, meaning the 12–volt power supply to the ASD relay will be de-activated by the PCM if:

- the ignition key is left in the ON position. This is if the engine has not been running for approximately 1.8 seconds.
- there is a crankshaft position sensor signal to the PCM that is lower than pre-determined values.

DIAGNOSIS AND TESTING - ASD AND FUEL PUMP RELAYS

The following description of operation and tests apply only to the Automatic Shutdown (ASD) and fuel pump relays. The terminals on the bottom of each relay are numbered. Two different types of relays may be used, (Fig. 2) or (Fig. 3).